Enhancing Riparian Management with AI and Multi-Platforms: Efficiency-improved Infrastructure and Environmental Monitoring

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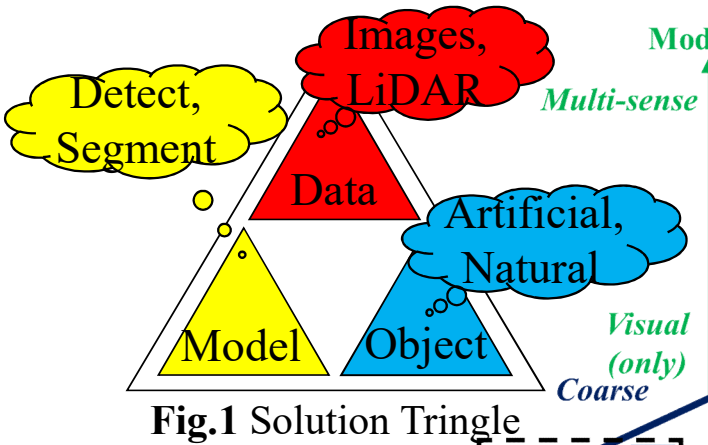

Fig.1 Solution Tringle

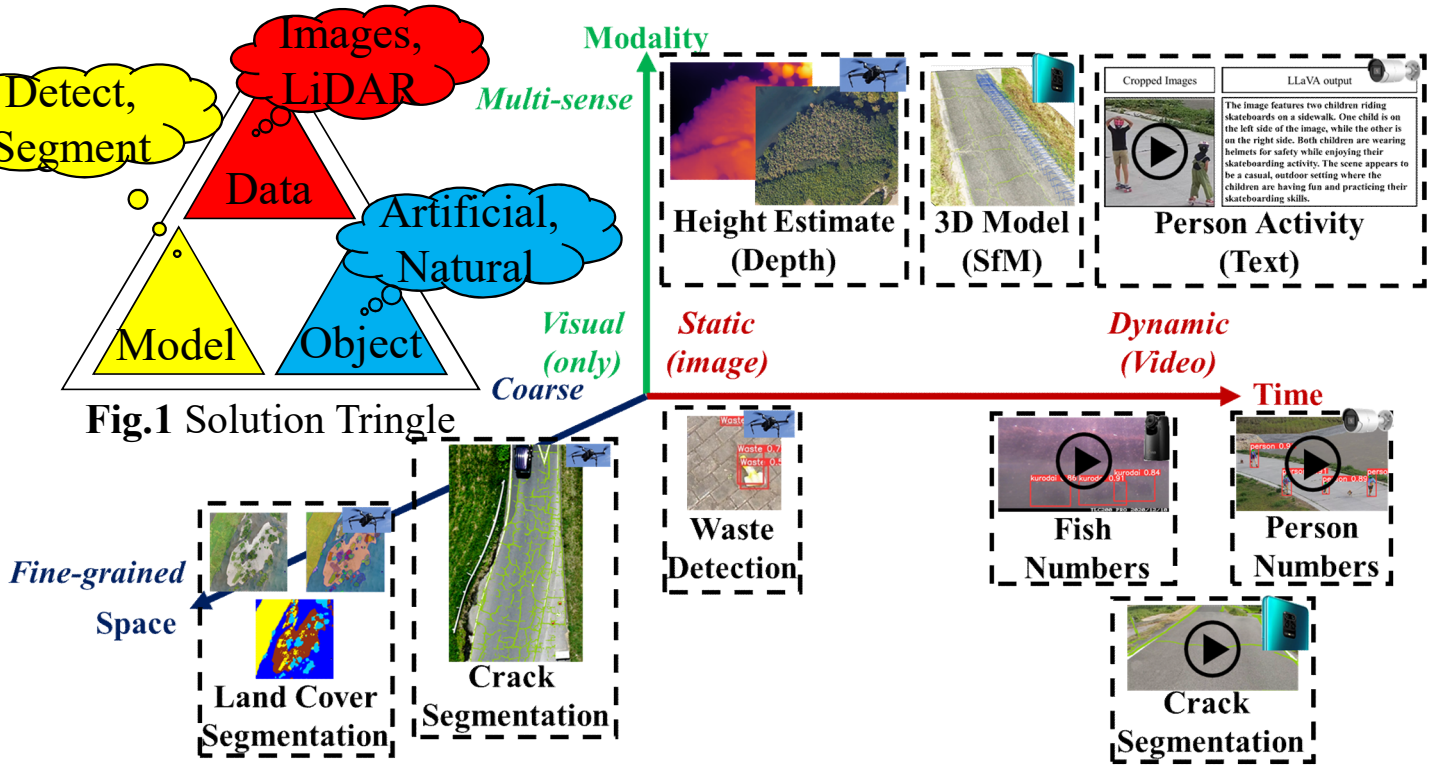

Fig.2 Overall Picture of the Research Themes from a Modality-Time-Space 3-D axis.

This research has pioneered several innovative methodologies that have significantly advanced the field of civil engineering, remote sensing and AI technology.

- **The development and application of the LiDAR-assisted DeepLabV3+ improved the accuracy of land cover classification tasks.**
- **The novel fusion methodology of aerial photography and LiDAR, utilizing a high contrast color scale, has expanded data features and improved performance.**
- **UAV-based detection tasks has shown promising results in asphalt-paved cracks and waste pollution detection.**
- **The creation of AI-based Generative Content (AIGC) has increased data availability and proven its potential in augmenting object detection tasks, particularly in waste pollution detection.**
- **Lastly, the enhancement of river patrolling tasks through AI-powered UAV technology has set a new standard for efficiency and effectiveness in environmental monitoring.**

Fig.3 Key-Words Cloud of the Research Themes.

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